Altimeter Setting Discrepancies at Landing: A Pre-registered Study of Business Jets and Airline Jets in Open ADS-B Data

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Abstract

An aircraft that lands with the wrong altimeter setting flies its barometric altitudes at the wrong height with no barometric indication of it. Two European studies have measured how often this happens; no published measurement was found for the United States, or one that compares types of operator. Modern aircraft broadcast the pressure setting selected on board. This study compared that broadcast setting with the consensus of other aircraft landing at the same airport within 30 minutes, at 119 US and European airports, in two pre-registered samples of 30 days in 2026 (about 714,000 and 730,000 landings). In the first sample business jets landed with a setting at least 2 hPa from the consensus more often than airline jets at the same airport on the same day (Mantel-Haenszel odds ratio 3.4). A second hypothesis, about Europe against the United States, was not supported, and review showed that the registered measure picked up a spurious switch to the standard setting in the final seconds on some European airliners. A corrected measure was then fixed in advance and tested on the second sample. The business-jet result replicated: odds ratio 4.0 (95% interval 3.5 to 4.7; 2.7 to 5.4 when whole operators are resampled), and 13.9 (7.9 to 23.7) for discrepancies of 5 hPa or more. Pooling both samples for description, the setting was 2 hPa or more from the reference on 2.0 per 1,000 landings, 5 hPa or more on 0.20, and 10 hPa or more (about 280 ft) on 0.06. The consensus was within 1 hPa of the airport weather report for 97% of landings, and the results were the same with the weather report as the reference. Nine in ten of the discrepancies are of 2 to 5 hPa and may not be errors in the operational sense. In exploratory analysis, about half of the large discrepancies recorded in Europe came from a small number of aircraft that supply the value by a second route and show the standard setting; with those removed, discrepancies of 5 hPa or more were two to three times as frequent in the United States as in Europe. Which altimeter the broadcast reflects is not defined, so on some business jets a discrepant broadcast may not mean the crew flew on a wrong setting.

Keywords: altimeter setting; QNH; ADS-B; business aviation; barometric altimetry; pre-registration; open data

1 Introduction

On 23 May 2022 an Airbus A320 on a satellite-based approach to Paris-Charles de Gaulle descended to about six feet above the ground about 0.9 NM short of the runway. The controller had passed an altimeter setting 10 hPa too high, the crew set it, and every barometric indication on board looked normal [1]. The European Union Aviation Safety Agency has issued and revised a safety bulletin on incorrect altimeter settings [2].

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Two published measurements of how often aircraft land on a wrong setting were found, both in Europe. The French civil aviation safety directorate compared the setting downlinked to radar with the setting displayed to controllers for 1,694,266 landings at six French airports from 2019 to 2023. It found a difference of 2 hPa or more at the runway threshold on 2,269 landings (about 1.3 per 1,000) and of 10 hPa or more on 387 (about 0.23 per 1,000). About 27% of the inconsistencies had the standard setting on board. The rate at Le Bourget, a business-aviation airport, was three times the rate at Charles de Gaulle, which the report attributed to "very different fleets" without further analysis [3]. A EUROCONTROL study built a detection algorithm on broadcast data and, as reported by France to an ICAO regional group, found 196 errors above 5 hPa in 747,353 flights at 378 European airports [4, 5]. The full text of that paper could not be obtained for this study.

No published rate was found for the United States, and no study was found that compares business aviation with airlines or one region with another. The United States differs from Europe in two relevant ways. Settings are given in inches of mercury, and the change from the standard setting to the local one is made at 18,000 ft, well before the approach, where in Europe it is made at a few thousand feet [3, 6].

This study uses open data to measure the rate at 119 airports in both regions and to answer two questions. Do business jets land with a discrepant setting more often than airline jets at the same airport on the same day? And does the rate differ between Europe and the United States, in particular for landings with the standard setting still selected?

2 Methods

2.1 Data and samples

The adsb.lol archive publishes each day's ADS-B traces under an open licence [7]. Two samples of 30 UTC days in 2026 were used. The first had been drawn at random for an earlier study of approach paths that never used the altimeter-setting field. The second was drawn at random from the 263 eligible days of 2026 not already used; one drawn day (27 March) could not be processed and was replaced by the first reserve day (25 March), as the plan provided. Airports (119, chosen in advance, 76 in the United States and 43 in Europe) and aircraft types (65 business-jet and 58 airline jet types) were fixed lists. Airline-type aircraft were kept only with an airline-format callsign. Airport elevations came from OurAirports [8].

2.2 Setting at landing

A landing is the last airborne point of a track aligned with a runway and followed by ground movement or no later climb. The setting at landing is the most frequent value of the selected barometric pressure setting recorded for the aircraft in the 120 seconds before that point [9]. A landing was excluded if three or more different values were recorded in that window, or if the value was outside 940 to 1060 hPa.

The recorded field mixes two sources. One is the ADS-B broadcast, in steps of 0.8 hPa. The other has 0.1 hPa resolution and is probably a Mode S register decoded by the receiver software. A value that is not on the 0.8 hPa grid can only have come from the second source. Section 3.3 reports results with and without such values.

2.3 Reference

The reference for a landing is the median setting at landing of other aircraft landing at the same airport within 30 minutes either side. It required at least five other aircraft, at least 60% of them within 1.0 hPa of their median. Landings without a reference were excluded.

The reference was checked against the most recent routine or special weather report (METAR) for the airport issued in the 70 minutes before the landing, from the Iowa State University archive [10]. Reports outside 940 to 1060 hPa and isolated spikes were removed (25 reports). The archive gives the setting in inches of mercury to two decimal places, so European values pass through a rounding of about 0.3 hPa. Palm Beach was missing from the weather data.

2.4 Discrepancies and exclusions

The difference is the setting at landing minus the reference, rounded to 0.1 hPa. A difference of 2 hPa or more (the primary threshold) is called a discrepancy, and one of 5 hPa or more a large discrepancy; 10 hPa was also used. The word "error" is avoided because a broadcast value that differs from the local setting is not always a mistake by the crew (Section 4). Because the broadcast moves in steps of 0.8 hPa, a threshold of 2 hPa against a reference built from other broadcasts is in effect 2.4 hPa. "Standard setting at landing" means a setting within 0.5 hPa of 1013.25 that was at least 5 hPa from the reference, among landings where the reference was at least 5 hPa from standard. Two exclusions were made without regard to class: settings matching station pressure at airports above 300 ft (use of QFE), and all landings by any aircraft that had a discrepancy on at least half of four or more landings (the persistent-discrepancy rule).

2.5 Statistical analysis

Business jets were compared with airline jets by the Mantel-Haenszel common odds ratio across strata of airport and UTC date, with a Cochran-Mantel-Haenszel test and a Robins-Breslow-Greenland interval. Because landings by one operator are not independent, intervals were also obtained by resampling whole operators (airline callsign prefix; aircraft without one treated singly) and whole airports. Rates are crude, with intervals from resampling airport-days. Regions were compared by the ratio of crude rates, with a p-value from permuting the region label across airports.

2.6 Pre-registration, deviations and review

The study was pre-registered twice, as dated commits in a public git repository (see Materials). No external registry was used, so the dates are not independently certified.

First sample (commit `be41aa0`, 11 November 2026, 13:18 UTC, before any altimeter-setting data from the study days was read). Two hypotheses: H1, that business jets land with a setting 2 hPa or more from the reference more often than airline jets; H2, that landing with the standard setting selected is more frequent in Europe than in the United States. The registered setting at landing was the *last* value in the 120-second window. Review found that measure flawed: on some aircraft, mostly European airliners, the second source shows the standard setting for a few seconds around touchdown after the correct setting had been shown throughout the approach, and a few aircraft broadcast values that jumped between impossible ones. The most-frequent-value measure described above was adopted in response. Appendix A gives the first-sample results exactly as registered.

Second sample (commit `7306d3b`, 11 November 2026, 14:39 UTC, before any of it was downloaded). The corrected measure and four hypotheses were fixed: business jets against airline jets for discrepancies of 2 hPa or more (R1), 5 hPa or more (R2), 2 hPa or more among aircraft with ten or more analysed landings (R3), and the standard setting at landing (R4), with Holm correction. R1 would count as replicated if supported with an odds ratio between 2 and 6. The frozen script was run once after all 30 days were processed.

The confirmatory results in this paper are therefore H1 and H2 on the first sample (Appendix A) and R1 to R4 on the second (Table 3). Everything else is exploratory, including all first-sample results on the corrected measure, every figure that pools the two samples, the separation of values by source, the operator-level intervals, the cleaning of weather reports and the comparison of regions. Pooled figures stack the analysed landings of the two samples, each prepared separately. Appendix B gives the results by sample.

All review of this paper has been by independent AI reviewers that had not seen the work being produced (see Acknowledgement). Their findings changed the measure and the analysis as described above. Every figure in this paper was computed by the author's analysis code; none is taken from a reviewer's report.

3 Results

Rates and descriptive figures in this section pool the two samples (60 days). The four hypothesis tests are reported on the second sample, where they were pre-registered, with the pooled estimate alongside. Appendix B gives every table by sample. The samples agree on every main result; the estimates that rest on few events differ more.

3.1 Samples

The two samples held 1,635,885 landings (Table 1). No setting was recorded on 48,169; 154 had three or more values in the window, 40 had implausible values, 5 matched QFE, and 131 were by aircraft excluded under the persistent-discrepancy rule. With landings lacking a reference also removed, 1,444,082 were analysed: 1,317,639 by 18,006 airline aircraft and 126,443 by 13,570 business jets. Far fewer business-jet landings could be analysed. The samples share most of their aircraft: 99.9% of airline and 96.6% of business-jet landings in the second were by aircraft that also appear in the first.

Table 1: Coverage by class and region, both samples.

Group	Landings	Setting recorded	Analysed
Airline jets, Europe	557,553	99.6%	93.8%
Airline jets, United States	889,929	97.2%	89.3%
Business jets, Europe	34,237	97.9%	75.0%
Business jets, United States	154,166	86.9%	65.3%

3.2 How often the setting differed

The setting differed from the reference by 2 hPa or more on 2.01 per 1,000 landings (95% interval 1.87 to 2.16), by 5 hPa or more on 0.20 (0.17 to 0.22), and by 10 hPa or more on 0.060 (0.048 to 0.073). Table 2 and Figure 1 give the rates by class and region. Nine in ten of the discrepancies of 2 hPa or more were smaller than 5 hPa, 82% of them at 2.4 or 3.2 hPa.

3.3 Values from the second source

Values off the 0.8 hPa grid were 1,096 of 548,444 analysed European landings (0.2%) and 14 in the United States, but they were 40 of the 88 European discrepancies of 5 hPa or more. Thirty-eight of the 40 were values at or near 1013. They came from 33 aircraft, with 23 of the 40 on A300-600, 737 Classic and 777 airframes and the rest on business jets. Eighteen of

Table 2: Landings per 1,000 with the setting at least the amount shown from the reference. Crude rates with event counts in brackets, both samples. "Grid only" rows keep only values on the 0.8 hPa broadcast grid (Section 3.3); US rows are unchanged by that restriction.

	Landings	2 hPa	5 hPa	10 hPa
Airline jets, Europe	522,755	0.28 (148)	0.11 (56)	0.033 (17)
grid only	521,808	0.20 (105)	0.057 (30)	0.013 (7)
Airline jets, United States	794,884	2.04 (1,622)	0.12 (98)	0.034 (27)
Business jets, Europe	25,689	1.83 (47)	1.25 (32)	0.51 (13)
grid only	25,540	1.29 (33)	0.70 (18)	0.43 (11)
Business jets, United States	100,754	10.8 (1,084)	0.97 (98)	0.30 (30)
All landings	1,444,082	2.01 (2,901)	0.20 (284)	0.060 (87)
grid only	1,442,972	1.97 (2,841)	0.17 (243)	0.052 (75)

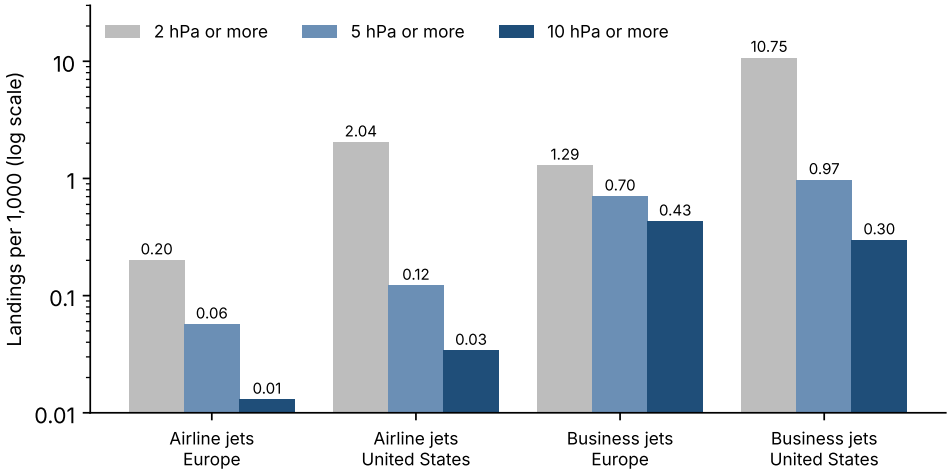


Figure 1: Landings per 1,000 with the setting at least the amount shown from the reference. Both samples, values on the broadcast grid, crude rates, logarithmic scale.

the 33 showed 1013 on at least half of their landings but made too few landings to trip the persistent-discrepancy rule; on the others it was occasional. A value of 1013 that appears only through the second source, on particular airframes, is far more likely to be a property of the avionics or the decoding than a crew landing on the standard setting. European rates are therefore given both ways in Table 2, and the grid-only figures are the better estimate. The grid test is approximate, since one 0.1 hPa value in eight falls on the grid by chance.

The European rates on grid values (0.25, 0.09 and 0.03 per 1,000 at 2, 5 and 10 hPa) are well below the two earlier European studies: about 1.3 and 0.23 at 2 and 10 hPa in the French study [3], and 0.26 at 5 hPa reported for EUROCONTROL [5]. The French study used the radar downlink, which may be the same second source, but this was not tested.

3.4 Business jets against airline jets

Within the same airport and date, business jets had about four times the odds of airline jets of landing with a setting 2 hPa or more from the reference, and about fourteen times the odds for 5 hPa or more in the second sample, eleven with the samples pooled (Table 3). All four pre-registered hypotheses were supported on the second sample after Holm correction

(each $p < .001$), and R1 met its replication criterion. The first-sample hypothesis H1, on the originally registered measure, was also supported (odds ratio 3.42; Appendix A).

Crude rates. Over all airports the rates of a discrepancy of 2 hPa or more were 8.9 per 1,000 for business jets and 1.34 for airline jets; of 5 hPa or more, 1.03 and 0.12. The crude ratios are larger than the odds ratios because business jets land more at airports where rates are higher for every class.

Table 3: Business jets against airline jets within airport and date. Mantel-Haenszel odds ratios; business-jet and airline events in informative strata in square brackets. For the second sample, where the tests were pre-registered, the first interval is the registered Robins-Breslow-Greenland one and the second comes from resampling whole operators (exploratory, and wider). The pooled estimate is exploratory, with operator-level intervals.

Outcome	Second sample	Both samples
R1: 2 hPa or more from reference	4.0 (3.5–4.7; 2.7–5.4) [377, 877]	3.7 (2.5–4.9) [719, 1,733]
R2: 5 hPa or more from reference	13.9 (8.9–21.8; 7.9–23.7) [64, 63]	11.1 (6.4–17.4) [113, 142]
R3: as R1, aircraft with 10+ landings	2.6 (2.1–3.2; 1.7–3.6) [135, 830]	2.6 (1.7–3.6) [265, 1,627]
R4: 1013 at landing, 5 hPa+ away	35 (16–78; 12–114) [34, 18]	34 (13–85) [61, 37]

Robustness. With the samples pooled, the R1 odds ratio was 3.6 in the United States and 7.2 (4.7 to 10.9) in Europe, and 3.7 on grid values only (R2 10.5). The second-sample sensitivity analyses are in Appendix B. With the reference built from airline aircraft only, it was 3.7 in the first sample and 4.0 in the second.

Regression models. A logistic model with airport fixed effects and standard errors clustered by operator, fitted to second-sample landings in informative strata, gave an odds ratio of 4.1 (2.9 to 5.6) for 2 hPa or more and 13.8 (8.3 to 23.2) for 5 hPa or more; pooled, the latter was 10.3 (6.6 to 16.0). Adding a random intercept for each operator gave 10.5 (8.0 to 13.8) for 5 hPa or more in the second sample. The random-intercept model could not be fitted for the 2 hPa outcome within the computing limits available.

Military and government flights. The airline class contained 308 landings with military or government callsigns or addresses, none with a discrepancy of 2 hPa or more. Removing them leaves every estimate unchanged.

By size of discrepancy. The odds ratio was 3.3 (2.2 to 4.3) for discrepancies of 2 to 5 hPa and 18 (7 to 39) for 10 hPa or more, the latter on 40 business-jet and 41 airline events. Ten of the 43 business-jet discrepancies of 10 hPa or more were non-standard values 30 hPa or more from the reference, against one for airline jets. Of the five in the second sample, two jumped to the implausible value within the last three minutes, two were one aircraft showing the same value on six landings, and one was a steady value about one inch of mercury from the reference, which is a plausible mis-set. Excluding such values from the second sample, R2 is 13.0 and the 10 hPa odds ratio 15.8.

By how often the aircraft was seen. The gap narrows among aircraft that landed often (Table 4). Among aircraft with twenty or more landings in a sample the odds ratio was 2.1 (1.3 to 3.2). Of the 43 business-jet discrepancies of 10 hPa or more, 34 were by aircraft with fewer than ten landings in the sample.

By operator, type and airport. Two fractional-ownership operators flew 26% of analysed business-jet landings. In the first sample, among the 60 airline callsign groups with at least 1,500 landings, rates ran from 0 to 14 per 1,000 with a median of 0.5, and several large carriers had none. Regional jets were highest among airline types (2.6 to 5.1 per 1,000). Business-jet

Table 4: Discrepancies of 2 hPa or more per 1,000 landings, by the number of landings the aircraft made in its 30-day sample. Both samples.

Landings by the aircraft	1	2–3	4–9	10–19	20 or more
Business jets	16.7	12.6	9.7	7.6	5.8
Airline jets	14.0	4.5	1.4	1.5	1.3

types with at least 1,000 landings ran from 3.1 to 16.2. The spread between operators is as wide as the gap between the classes. Rates for both classes rose with the share of business jets at the airport, where the reference is also noisier. Aircraft with a discrepancy in the first sample had a higher rate in the second: 14.3 against 8.5 per 1,000 for business jets, 2.9 against 1.2 for airline jets.

3.5 Standard setting at landing

About half of business-jet discrepancies of 5 hPa or more had 1013 selected (67 of 130). Table 5 gives the counts. Within airport and date, business jets had far higher odds than airline jets (Table 3, R4). Most European airline events were second-source values; on grid values only the pooled odds ratio is 50 (22 to 118), on 47 business-jet and 18 airline events.

Table 5: Landings with 1013 selected and the reference at least 5 hPa away: events / eligible landings, both samples.

Group	All values	Grid only
Airline jets, Europe	33 / 330,952	9 / 330,389
Airline jets, United States	11 / 435,800	10 / 435,798
Business jets, Europe	24 / 15,603	10 / 15,508
Business jets, United States	43 / 55,310	43 / 55,305

The traces of all 35 business-jet events in the second sample were examined. Broadcast values were available for a median of six minutes before landing. On 31 the value was 1013 throughout that time, one changed to 1013 during it, and three showed other patterns. All 35 were different aircraft, 34 of which showed a non-standard value on another landing, and they were of 22 types, old and recent. They are not stuck transponders and not the touchdown artefact, but the traces cannot show whether the crew's own altimeter was also at 1013.

The first-sample hypothesis H2, that the standard setting at landing is more frequent in Europe, was not supported as registered (rate ratio 1.97, $p = .072$), and the apparent European excess was the touchdown artefact: on the corrected measure the ratio was 0.95 ($p = .85$).

3.6 The United States against Europe

On grid values, discrepancies of 2 to 5 hPa were seventeen times as frequent in the United States as in Europe (2.80 against 0.16 per 1,000), and discrepancies of 5 hPa or more were 2.5 times as frequent (0.22 against 0.088; resampling airports, 1.6 to 4.1; significant in each sample). On all values the second ratio was 1.4 (0.95 to 2.1; $p = .11$). The comparison is exploratory and is not adjusted for the mix of traffic.

3.7 The reference against weather reports

The consensus reference was within 1.0 hPa of the latest weather report for 97.4% of 1,432,931 landings (97.0% in Europe, 97.7% in the United States) and within 1.7 hPa for 99.9%. Agreement was lower at airports where business jets were over 60% of traffic (94.3%).

With the weather report as the reference in place of the consensus, which brings in about 40,000 business-jet landings that had too little other traffic for a consensus, the odds ratio was 3.8 (3.4 to 4.2) at the matched threshold of 2.4 hPa and 12.4 (9.3 to 16.6) at 5 hPa. The two references give the same answer. (On the second sample alone these are 4.0 and 13.1. The frozen second-sample script, which did not remove corrupt reports and used a nominal 2.0 hPa threshold that admits two-step differences, gave 2.8 and 9.4.) Measured against weather reports, landings without a consensus reference had higher discrepancy rates in both classes (business jets 9.7 against 6.7 per 1,000, airline jets 2.6 against 0.9), so the rates in Table 2 are probably too low.

3.8 Other observations

- **Small discrepancies in the United States.** In the first sample, airline ones bunched in the first ten minutes of each hour (177, against 101 to 127 in the other ten-minute periods), which fits the hourly weather update; business-jet ones were flat (74 to 87). In the second sample only 24% of airline and 11% of business-jet small discrepancies matched a setting published for the destination in the previous two hours. Across both samples, 26% of business-jet and 20% of airline small discrepancies matched, within 0.5 hPa, the current setting at another study airport within 100 NM, against 9% and 6% for a mirror-image value used as a chance baseline. Only the 118 study airports with weather data could be tested, about three per landing, so the true share is likely higher.
- **Direction (first sample).** Discrepancies of 2 hPa or more were evenly split. Of discrepancies of 5 hPa or more, 50 were set too high, which puts the aircraft lower than indicated, and 94 too low.
- **One-off and steady.** Of 1,131 business-jet discrepancies, 1,113 showed a single value throughout the last two minutes and 1,095 were by aircraft that showed other values on other landings. Only 51 of 1,078 aircraft had more than one in a sample.
- **Paris.** Le Bourget had too few analysable landings to test the French threefold difference.

4 Discussion

What is established. Compared at the same airport on the same day, the odds that a business jet lands with a broadcast altimeter setting 2 hPa or more from that of other aircraft are about four times those for an airline jet. The result was confirmed in a second pre-registered sample of new days, is the same when the airport weather report is the reference, and holds with allowance for landings by the same operator (interval 2.7 to 5.4). It is not produced by a few faulty transponders.

Most of it is small. Nine in ten of the discrepancies are between 2 and 5 hPa, mostly 2.4 or 3.2 hPa, equivalent to about 65 to 90 ft. What they are is not established. In the United States, the bunching of airline ones after the hourly update (first sample) points to settings not yet renewed, but only a minority in either class matched a recent setting for the

destination. In the United States a setting taken from a nearby airport accounts for about a seventh of them beyond chance, and probably more, since only a few nearby stations could be tested; a setting from a station within 100 NM is legitimate en route. These are best described as discrepancies from the local setting, not errors in the sense of the Paris incident.

The large discrepancies are few. Discrepancies of 5 hPa or more occurred on about 1 business-jet landing per 1,000 against about 0.1 for airline jets. The ratio is large and was confirmed, and it is the safety-relevant finding. It rests on 130 business-jet events in 60 days, about half of them with 1013 selected. At 10 hPa most came from aircraft seen fewer than ten times, and about a fifth look like corrupt values.

The broadcast may not be the altimeter the crew flew on. The standards define the field as the setting selected by the crew but not which side it comes from [11]. On some light jets flown by one pilot the transponder may take its setting from the unused side, and such aircraft may lack a warning when the two sides disagree; this possibility was raised in review and could not be confirmed from any public document. The data cannot exclude that reading, and the spread of 1013 events across 22 types gives it no clear support either. On this reading some business-jet "standard at landing" events are an unset second altimeter and not a crew flying on the wrong setting. That would still mean no cross-check between the two sides was made.

The United States against Europe. Once the second-source values are set aside, discrepancies of 5 hPa or more were two to three times as frequent in the United States as in Europe, in both samples. This does not support treating the problem as a European one that the 18,000 ft transition altitude prevents. The comparison is exploratory and crude, and the European rates here are well below both earlier studies, which may have included the same second-source values.

Consequences. A 10 hPa error is about 280 ft. On an approach with ground-based vertical guidance the glidepath is unaffected, but the decision altitude, the altitude check on the glidepath, step-down and missed-approach altitudes are all displaced. On an approach using barometric vertical guidance the whole path is displaced [1]. Radio altimeters, terrain warning using satellite height and controller warnings remain as defences. The study does not know which type of approach was flown.

5 Limitations

- **Review.** The paper has been reviewed only by AI reviewers. The pre-registrations, analysis and review all took place within two days, and the commit times are not independently certified.
- **The measure is not direct.** The recorded value combines two sources that can disagree, and which altimeter either reflects is unknown. Values from the second source could be separated only approximately. The measure was chosen after seeing a problem in the first sample, then fixed in advance for the second.
- **Inferred reference.** The reference is the consensus of other aircraft. A wrong value given to every aircraft would not be detected. It agreed with weather reports within 1 hPa for 97% of landings.
- **Coverage and selection.** Only 65% of US and 75% of European business-jet landings could be analysed, against about 90% for airline jets. The missing landings are aircraft that do not broadcast a setting, which is optional in the United States, and landings with too little other traffic. Across the 44 business-jet types with at least 1,000 landings,

types with lower broadcast coverage had higher discrepancy rates among the aircraft that did broadcast (Spearman correlation -0.54), so the rates here may understate the true business-jet rate.

- **Exposure.** Airline aircraft averaged about 37 landings per sample and business jets about 5.5. The per-landing comparison is dominated by rarely seen business jets, and the gap is smaller among frequently seen aircraft.
- **Lag counted as discrepancy.** A crew holding a correct but older value is counted at the 2 hPa threshold.
- **Classes.** The airline class includes cargo flights. Military and government flights identified by callsign or address were too few to matter (308 landings), but the list used was a judgement. Operator type is inferred from aircraft type and callsign. Test and training flights are included.
- **No operational context.** Approach type, weather, minima, crew composition and what ATC said are unknown.
- **Sample.** Sixty days in 2026 at 119 airports chosen by judgement. The second sample has new days but the same airports and mostly the same aircraft. It confirms that the result is stable over time and was not produced by choices made after seeing the data. It does not show that the result would hold for other airports or fleets.
- **Prior work not fully read.** The EUROCONTROL paper was seen only as an abstract and through secondary reports, and its figures quoted here come from a secondary source. It may contain breakdowns that this paper describes as not found. No systematic search was made for unpublished monitoring by air navigation service providers.

6 Conclusion

In open ADS-B data from 119 airports, a broadcast altimeter setting at least 2 hPa from the local one was present on about 2 landings per 1,000, and one at least 5 hPa away on about 2 per 10,000. At the same airport on the same day, business jets had about four times the odds of airline jets of the first. That was predicted before any data was read and confirmed on a second sample. For the second, the odds were about fourteen times as high in the second sample, where that hypothesis was registered after the first sample had been seen. In exploratory analysis, large discrepancies were not confined to Europe. Whether the business-jet excess reflects the altimeter the crew flew on, a second altimeter left unset, or differences in how settings are obtained cannot be settled from broadcast data, and needs avionics evidence for the main types and operator flight data. The standards leave the source of the broadcast setting to the manufacturer and installer, and no public document was found that states it for any of the main business-jet avionics suites.

A First-sample results as pre-registered

These results use the registered measure, the last value in the 120-second window, and the frozen code, whose hashes match the pre-registration. They are superseded by the main text for the reasons given in Section 2.6.

Sample. A setting was recorded in the last two minutes on 97.0% of the 809,959 landings. After exclusions (27 implausible values, 3 landings on QFE, and 89 landings by 9 aircraft, 8 of them business jets, under the persistent-discrepancy rule), 713,835 landings were analysed:

650,307 by 17,617 airline aircraft and 63,528 by 11,618 business jets. Stratum dates span 31 UTC dates because three landings fell just after midnight.

H1. Within the same airport and date, the odds ratio of a discrepancy of 2 hPa or more for business jets against airline jets was 3.42 (95% interval 2.94 to 3.98; resampling airports, 2.77 to 4.27; Holm-adjusted $p < .001$; 684 informative strata). The crude rates were 8.53 per 1,000 (542 of 63,528) and 1.43 per 1,000 (930 of 650,307). H1 is supported.

H2. "Standard left set" was registered as a setting within 0.5 hPa of 1013.25 that differed from the reference by 2 hPa or more, among landings where the reference was at least 3 hPa from standard. It was recorded on 71 of 188,811 eligible landings in Europe (0.38 per 1,000) and 58 of 303,311 in the United States (0.19 per 1,000): rate ratio 1.97 (0.99 to 3.49; permutation $p = .072$). H2 is not supported.

Rates. The registered measure gave 2.06 discrepancies of 2 hPa or more per 1,000 landings, 0.26 of 5 hPa or more and 0.083 of 10 hPa or more.

Effect of the correction. The corrected setting differed from the registered one on 1,784 first-sample landings, and 80 landings were excluded as unstable. On the corrected measure the H1 odds ratio was 3.46 (2.97 to 4.02).

Other deviations (both samples). The rate tables by aircraft type use lower cut-offs than the frozen script. In the frozen replication script the two rate labels in the United States against Europe output are transposed; the ratio and p-value are unaffected. One trace file on the reserve day of the second sample could not be read.

B Results by sample

Tables 6 to 10 give the figures for each sample separately. The second-sample tests in Table 8 are the pre-registered ones.

Table 6: Coverage by class and region.

	First sample			Second sample		
Group	Landings	Recorded	Analysed	Landings	Recorded	Analysed
Airline jets, Europe	275,237	99.6%	93.6%	282,316	99.6%	93.9%
Airline jets, United States	439,679	97.2%	89.3%	450,250	97.2%	89.3%
Business jets, Europe	17,416	97.9%	74.9%	16,821	97.8%	75.1%
Business jets, United States	77,627	86.9%	65.0%	76,539	86.9%	65.7%

Other per-sample figures. Discrepancies of 5 hPa or more, United States over Europe, on grid values: 2.2 in the first sample ($p = .002$) and 2.8 in the second ($p < .001$); on all values 1.4 ($p = .15$) and 1.3 ($p = .22$). Consensus reference within 1.0 hPa of the weather report: 97.4% and 97.5%. Weather report as reference: odds ratios 3.5 and 4.0 at 2.4 hPa, 11.5 and 13.1 at 5 hPa. Second-source values among European discrepancies of 5 hPa or more: 17 of 44 and 23 of 44. Business-jet discrepancies of 10 hPa or more by aircraft with fewer than ten landings: 19 of 22 and 15 of 21. In the second sample, the R1 odds ratio was 3.9 in the United States and 7.4 in Europe, 3.8 with the hour added to the strata, 4.8 with a reference from ten or more aircraft with over 95% agreement, 3.4 for discrepancies of 2 to 5 hPa, 20 (9 to 44) for 10 hPa or more, 3.1 among aircraft with four or more landings and 2.2 among those with twenty or more. The registered intervals from resampling whole airports were 3.2 to 5.3 for R1 and 9.4 to 21.4 for R2. Second-sample exclusions were 74 landings

Table 7: Landings per 1,000 with the setting at least the amount shown from the reference. Crude rates, first sample / second sample. "Grid only" rows keep only values on the 0.8 hPa broadcast grid ; US rows are unchanged by that restriction.

	2 hPa	5 hPa	10 hPa
Airline jets, Europe	0.26 / 0.31	0.10 / 0.11	0.023 / 0.041
grid only	0.19 / 0.21	0.070 / 0.045	0.012 / 0.015
Airline jets, United States	2.05 / 2.03	0.14 / 0.10	0.038 / 0.030
Business jets, Europe	1.69 / 1.98	1.30 / 1.19	0.54 / 0.47
grid only	1.08 / 1.51	0.69 / 0.72	0.46 / 0.40
Business jets, United States	10.3 / 11.3	0.87 / 1.07	0.30 / 0.30
All landings	1.98 / 2.04	0.20 / 0.19	0.060 / 0.060
grid only	1.94 / 2.00	0.18 / 0.16	0.055 / 0.049

Table 8: Business jets against airline jets within airport and date. Mantel-Haenszel odds ratios; business-jet and airline events in informative strata in square brackets. The first interval is the Robins-Breslow-Greenland one; the second, for the second sample, comes from resampling whole operators and is the one to rely on. Second-sample tests were pre-registered; first-sample figures are exploratory.

Outcome	First sample	Second sample
R1: 2 hPa or more from reference	3.5 (3.0–4.0) [342, 856]	4.0 (3.5–4.7; 2.7–5.4) [377, 877]
R2: 5 hPa or more from reference	8.6 (5.2–14.2) [49, 79]	13.9 (8.9–21.8; 7.9–23.7) [64, 63]
R3: as R1, aircraft with 10+ landings	2.6 (2.1–3.1) [130, 797]	2.6 (2.1–3.2; 1.7–3.6) [135, 830]
R4: 1013 at landing, 5 hPa+ away	32 (13–77) [27, 19]	35 (16–78; 12–114) [34, 18]

with three or more values, 20 implausible values, 3 on QFE and 53 by 6 business jets under the persistent-discrepancy rule; strata span 33 UTC dates because three landings fell just after midnight. In the first sample, on the registered measure, every pre-registered sensitivity analysis of H1 gave between 3.1 and 5.1 (`results/secondary.json`, which also holds the registered exploratory analyses not reported here).

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Competing interests. The author is the founder of CitadelAero, which builds AI tools for aviation, and is employed as a business-jet pilot.

Disclaimer. The views expressed are the author's own and do not represent those of any employer or regulator. Nothing here is operational, legal or regulatory advice.

Data and code availability. The raw data are public: ADS-B traces from the adsb.lol archive, weather reports from the Iowa Environmental Mesonet, airport data from OurAirports. All code and result files are in the repository below. The processed per-landing tables are too

Table 9: Discrepancies of 2 hPa or more per 1,000 landings, by the number of landings the aircraft made in the sample (first sample / second sample).

Landings by the aircraft	1	2–3	4–9	10–19	20 or more
Business jets	13.5 / 19.9	10.8 / 14.4	10.4 / 9.1	7.2 / 8.0	5.2 / 6.4
Airline jets	16.4 / 11.7	3.0 / 6.0	1.5 / 1.3	1.5 / 1.4	1.3 / 1.3

Table 10: Landings with 1013 selected and the reference at least 5 hPa away: events / eligible landings. The last column keeps only values on the broadcast grid, second sample.

Group	First sample	Second sample	Second, grid only
Airline jets, Europe	15 / 158,123	18 / 172,829	2 / 172,519
Airline jets, United States	6 / 218,843	5 / 216,957	5 / 216,956
Business jets, Europe	14 / 8,040	10 / 7,563	4 / 7,517
Business jets, United States	18 / 27,437	25 / 27,873	25 / 27,871

large for the repository and are available from the author on request.

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Ethics. The study uses only publicly broadcast aircraft data and involves no human participants. Individual aircraft and operators are not named in the results, apart from aircraft types.

Materials. The pre-registration, code, results and audit record for this paper are at `https://github.com/JamesLeCuirot/Research/tree/prereg/08-wrong-altimeter-setting/papers/08-wrong-altimeter-setting`.

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